Materials and Methods: Fluorescence-Based Enzyme Kinetics Analysis

1. Data Acquisition and Visualization

1.1 Raw Fluorescence Data (Experimental_Results)

Raw fluorescence intensity (RFU) was measured as a function of time. Data were plotted without fitting to visualize the time-dependent fluorescence signal during enzymatic substrate cleavage.

1.2 Time-Course Fluorescence Curve Fitting

For substrate concentration series, fluorescence time courses were fitted to an exponential association model as described in **Equation (1)**:

$$F(t) = F_0 + (F_{max} - F_0)(1 - e^{-kt}) \quad \textbf{(Equation 1)}$$

where $F(t)$ is the fluorescence intensity at reaction time t (RFU), F_0 is the fluorescence at $t = 0$, and k is the apparent first-order rate constant (min^{-1}). F_{max} is the fluorescence intensity of the experimentally observed plateau, determined as the mean of the data points where $dF/dt \approx 0$ at $t \geq 3\tau$, and treated as a fixed parameter during curve fitting. F_{max} is distinct from F_∞ — the fluorescence intensity corresponding to 100% substrate cleavage, obtained from a complete-digestion control — in that F_{max} reflects the ceiling reached under a specific experimental condition and may fall below F_∞ if the reaction did not reach full completion within the observation window. Accordingly, F_{max} is used as the normalization reference throughout Sections 1–2 (Equations 1, 2, 4, 8, and 12), whereas F_∞ is reserved for calculation of the true fraction cleaved α in Section 3 (Equations 13–15 and 19).

To relate the rate constant k to the experimentally determined initial velocity v_0, we take the time derivative of **Equation (1)**:

$$dF/dt = (F_{max} - F_0) \cdot k \cdot e^{-kt}$$

At $t = 0$, the instantaneous rate of fluorescence increase equals the initial velocity v_0:

$$v_0 = [dF/dt]_{t=0} = (F_{max} - F_0) \cdot k$$

Rearranging this relationship yields **Equation (2)**:

$$k = v_0 / (F_{max} - F_0) \quad \textbf{(Equation 2)}$$

For enzyme concentration series at saturating substrate, the reaction exhibited pseudo-zero-order kinetics and was fitted to a linear model, shown in **Equation (3)**:

$$F(t) = F_0 + v_{0,raw}t \quad \text{(Equation 3)}$$

where t is the reaction time (min) and $v_{0,\text{raw}}$ is the initial velocity in raw fluorescence units (RFU/min).

1.3 Normalized Fluorescence Analysis

To compare reaction progress across different conditions, fluorescence data were normalized using **Equation (4)**:

$$F_{norm} = (F - F_0) / (F_{max} - F_0) \quad \text{(Equation 4)}$$

where F_{norm} is the normalized fluorescence intensity scaled from 0 to 1, obtained by referencing the *observed plateau* F_{max} rather than the complete-cleavage reference F_∞. Accordingly, $F_{\text{norm}} = 1$ indicates that the signal has reached the experimentally observed plateau, but does not necessarily correspond to full substrate cleavage (see note above). The initial linear region of the fluorescence time course, used to determine $v_{0,\text{raw}}$, was defined as the time window over which $F_{\text{norm}}(t) \leq 0.1$ (i.e., <10% substrate conversion). Within this range, substrate depletion is negligible and $F(t)$ increases linearly with time, validating the use of **Equation (3)**. For a fixed time-window approach, linear regression was also applied over predefined intervals (e.g., 0–30 s, 0–1 min, 0–2 min, 0–3 min), and the window yielding the highest linearity (R^2) was selected. Normalized data were then fitted over the full time course to **Equation (5)**:

$$F_{norm}(t) = 1 - e^{-k_{obs} \cdot t} \quad \text{(Equation 5)}$$

where k_{obs} is the observed rate constant from exponential fitting (min^{-1}), serving as the primary kinetic parameter. The initial linear phase of the normalized curve is described by the tangent **Equation (6)**:

$$y = k_{obs} \cdot t \quad \text{(Equation 6)}$$

where the slope k_{obs} represents the initial normalized velocity ($v_{0,\text{norm}} = k_{\text{obs}}$, in units of fraction/min or %/min). Converting back to raw fluorescence units, the tangent at $t = 0$ becomes:

$$F(t) \approx F_0 + k_{obs} \cdot (F_{max} - F_0) \cdot t = F_0 + v_{0,raw} \cdot t$$

which is consistent with **Equation (3)**, confirming that $v_{0,\text{raw}} = k_{\text{obs}} \cdot (F_{\text{max}} - F_0)$ regardless of whether the initial velocity is extracted from a direct linear fit or derived from the normalized exponential fit.

The time constant τ was calculated from **Equation (7)**:

$$\tau = 1 / k_{obs} \quad \text{(Equation 7)}$$

where τ is the time constant (min), representing the time required to reach approximately 63% of the maximum normalized response (i.e., $F_{\text{norm}}(\tau) = 1 - e^{-1} \approx 63.2\%$). The τ-based framework provides two

practical criteria for data analysis:

- **Initial velocity region (t ≤ 0.1τ):** At $t = 0.1\tau$, $F_{norm}(0.1\tau) = 1 - e^{-0.1} \approx 9.5\%$, confirming that the <10% substrate conversion threshold used to define the linear region in **Equation (3)** is mathematically equivalent to restricting the fit to $t \leq 0.1\tau$. Within this window, substrate depletion is negligible and the fluorescence signal increases linearly with time.
- **Exponential fitting range (t ≤ 3τ):** At $t = 3\tau$, $F_{norm}(3\tau) = 1 - e^{-3} \approx 95\%$, meaning the curve has effectively reached plateau and all information-rich curvature is captured within this window. Data points beyond 3τ lie within the flat plateau region and carry no additional kinetic information; however, they may be included in the fit without introducing bias, as they do not distort the exponential parameter estimates.

For comparison with measurements in the original fluorescence scale, the initial velocity in raw units ($v_{0,raw}$, RFU/min) was derived from k_{obs} using **Equation (8)**:

$$v_{0,raw} = k_{obs} \cdot (F_{max} - F_0) \quad \textbf{(Equation 8)}$$

This transformation allows consistent comparison across different experimental runs by deriving all velocity measurements from the normalized rate constant k_{obs}, which is less sensitive to variations in instrument signal intensity.

1.4 Michaelis-Menten Kinetic Analysis

Initial velocities ($v_{0,raw}$) determined from multiple substrate concentrations were fitted to the Michaelis-Menten equation, **Equation (9)**:

$$v_{0,raw} = V_{max} \cdot [S] / (K_m + [S]) \quad \textbf{(Equation 9)}$$

where V_{max} is the maximum reaction velocity at saturating substrate concentration, K_m is the Michaelis constant (substrate concentration at $V_{max}/2$, µM), and $[S]$ is the substrate concentration (µM).

1.5 Enzyme Concentration Dependence

When substrate concentration is held constant, the linear dependence of initial velocity on enzyme concentration can be derived directly from the Michaelis–Menten equation (**Equation 9**), which takes the general form:

$$v_{0,raw} = V_{max} \cdot [S] / (K_M + [S])$$

In enzyme kinetics, the maximum velocity V_{max} is not a fundamental constant but depends on the total enzyme concentration $[E]_T$ through the relation $V_{max} = k_{cat} \cdot [E]_T$, where k_{cat} is the catalytic turnover number (s^{-1} or min^{-1}). Substituting this into the Michaelis–Menten equation gives:

$$v_{0,raw} = k_{cat} \cdot [E]_T \cdot [S] / (K_M + [S])$$

When $[S]$ is held constant throughout the experiment, the entire factor $k_{cat} \cdot [S] / (K_M + [S])$ becomes a constant with respect to enzyme concentration. Collecting this constant as the proportionality slope explicitly reveals the linear dependence on $[E]_T$:

$$v_{0,raw} = \underline{k_{cat} \cdot [S]_{fixed} / (K_M + [S]_{fixed})} \cdot [E]_T$$

Thus, at a given fixed substrate concentration, a plot of $v_{0,raw}$ versus $[E]_T$ is expected to be linear through the origin, with a slope equal to $k_{cat}[S]_{fixed} / (K_M + [S]_{fixed})$. In practice, a non-zero intercept may arise from background fluorescence or baseline offset, so the data were fitted using **Equation (10)**:

$$v_{0,raw} = slope \cdot [E] + intercept \qquad \textbf{(Equation 10)}$$

where $[E]$ is the enzyme concentration (nM or µg/mL) and the slope corresponds to $k_{cat}[S] / (K_M + [S])$ under the fixed substrate condition. A supplementary analysis using only the lowest three enzyme concentrations (for quality control purposes) was performed with **Equation (11)**:

$$v_{0,raw} = a \cdot [E] + b \qquad \textbf{(Equation 11)}$$

where a and b are linear regression parameters.

2. Data Transformation

2.1 Back-Transformation to Original Scale

Fitted parameters from normalized space were converted back to original fluorescence units using **Equation (12)**:

$$F(t) = F_{norm}(t) \cdot (F_{max} - F_0) + F_0 \qquad \textbf{(Equation 12)}$$

3. Reaction Progress and Kinetic Modeling

3.1 Fraction Cleaved

The fraction of substrate cleaved $\alpha(t)$ was calculated from **Equation (13)**:

$$\alpha(t) = (F_t - F_0) / (F_\infty - F_0) \qquad \textbf{(Equation 13)}$$

where $\alpha(t)$ is the fraction of substrate cleaved (dimensionless, 0–1), F_t is the observed fluorescence at time t, and F_∞ is the fluorescence intensity upon complete substrate cleavage, determined from a separate complete-digestion control or extrapolated from kinetic model fitting. Because F_∞ — not F_{max} — is used as the denominator, $\alpha(t)$ correctly represents the absolute fraction of substrate cleaved even when the

reaction has not yet reached full completion within the observation window. Values were constrained to the range [0, 1].

3.2 Progress Curve Fitting

Full reaction time courses were fitted using F_∞ (the complete-cleavage reference) as the asymptotic parameter rather than F_{max}, so that the fitted amplitude directly reports the extent of reaction at saturation. Either a three-parameter exponential model, **Equation (14)**:

$$F(t) = F_0 + (F_\infty - F_0)(1 - e^{-k_{obs} \cdot t}) \quad \textbf{(Equation 14)}$$

or a two-parameter model assuming $F_0 = 0$, **Equation (15)**:

$$F(t) = F_\infty(1 - e^{-k_{obs} \cdot t}) \quad \textbf{(Equation 15)}$$

3.3 Kinetic Model A: Substrate Depletion (First-Order Approximation)

Under conditions where substrate concentration greatly exceeds K_m, the reaction follows pseudo-first-order kinetics described by **Equation (16)**:

$$\alpha(t) = 1 - e^{-k_{obs} \cdot t} \quad \textbf{(Equation 16)}$$

where the observed rate constant relates to enzyme concentration through **Equation (17)**:

$$k_{obs} = (k_{cat}/K_M) \cdot [E] \quad \textbf{(Equation 17)}$$

where k_{cat}/K_M is the specificity constant (enzyme efficiency metric).

3.4 Model A: Full Michaelis-Menten Integration

For comprehensive modeling, the substrate depletion was described by the differential equation, **Equation (18)**:

$$d\Gamma/dt = -v, \text{ where } v = k_{cat} \cdot [E] \cdot \Gamma / (K_M + \Gamma) \quad \textbf{(Equation 18)}$$

where Γ represents the remaining substrate concentration at time t. The fraction cleaved was calculated using **Equation (19)**:

$$\alpha = 1 - \Gamma_t / \Gamma_0 \quad \textbf{(Equation 19)}$$

where Γ_t is the substrate concentration at time t and Γ_0 is the initial substrate concentration, with α constrained to [0, 1].

4. Summary Table

A summary of all kinetic plots and their corresponding mathematical models is provided below.

Plot Type / Analysis	Key Equation	Ref.
Time-Course Fluorescence (Interpolated)	$F(t) = F_0 + (F_{max} - F_0)(1 - e^{-kt})$	**Eq. (1)**
Linear Enzyme Kinetics (Saturating [S])	$F(t) = F_0 + v_{0,raw}t$	**Eq. (3)**
Normalized Fluorescence	$F_{norm}(t) = 1 - e^{-k_{obs}t}$	**Eq. (5)**
Initial Linear Tangent (Normalized)	$y = k_{obs} \cdot t$	**Eq. (6)**
Raw Initial Velocity (from k_{obs})	$v_{0,raw} = k_{obs} \cdot (F_{max} - F_0)$	**Eq. (8)**
Initial Velocity vs. Substrate ([S])	$v_{0,raw} = V_{max}[S] / (K_m + [S])$	**Eq. (9)**
Initial Velocity vs. Enzyme ([E])	$v_{0,raw} = \text{slope} \cdot [E] + \text{intercept}$	**Eq. (10)**
Fraction Cleaved Calculation	$\alpha(t) = (F_t - F_0) / (F_\infty - F_0)$	**Eq. (13)**
First-Order Depletion Model	$\alpha(t) = 1 - e^{-k_{obs}t}$	**Eq. (16)**
Rate-Enzyme Concentration Relationship	$k_{obs} = (k_{cat}/K_M) \cdot [E]$	**Eq. (17)**

Note: All initial velocity measurements are derived from the primary kinetic parameter k_{obs} obtained from normalized exponential fitting (**Equation 5**), ensuring consistency across different experimental runs and minimizing the influence of instrument-specific signal variations. Early linear-slope velocity measurements may be used for quality control purposes but are not the primary reported values.